

STATISTICAL BUSINESS ANALYSIS

1. Project Overview

Project Title

Statistical Business Analysis

Objective

The objective of this project is to analyze business sales data using statistical methods and hypothesis testing. The analysis focuses on understanding sales patterns, comparing sales between regions, and determining whether observed differences are statistically significant.

Statistical techniques are used to support data-driven business interpretation and provide meaningful insights from the available sales data.

2. Dataset Description

The analysis uses business sales data containing information related to sales transactions.

The main variables used in the analysis include:

- Date
- Product
- Quantity
- Price
- Customer ID
- Region
- Total Sales

The `Total_Sales` variable is used as the primary measure for the statistical analysis.

The dataset was cleaned before analysis by converting numerical fields to appropriate numeric data types and removing records with missing values in the required numerical columns.

3. Statistical Analysis

Descriptive and inferential statistical techniques were applied to understand the characteristics of the business data.

The analysis included:

- Descriptive statistics
- Data cleaning
- Sales analysis
- Regional comparison
- Hypothesis testing
- Statistical significance analysis

The analysis was performed using Python libraries including Pandas, NumPy, SciPy, Matplotlib, Seaborn, and Scikit-learn.

4. Hypothesis Testing

A significance level of 0.05 was used for all hypothesis tests.

4.1 One-Sample T-Test

The one-sample t-test was used to determine whether the population mean of Total_Sales differs from the benchmark value of 97,955.500.

Results:

- Benchmark = 97,955.500
- t-statistic = 2.565
- p-value = 0.012
- Significance level = 0.050
- Decision = Reject H0

Interpretation

Since the p-value (0.012) is less than the significance level (0.05), the null hypothesis is rejected.

The result provides statistically significant evidence that the population mean of Total_Sales differs from the benchmark value of 97,955.500.

4.2 Independent-Samples Welch T-Test

A Welch independent-samples t-test was performed to compare the mean Total_Sales between the North and South regions.

Results:

- North mean = 142,272.679
- South mean = 138,438.963
- t-statistic = 0.130
- p-value = 0.897
- Significance level = 0.050
- Decision = Fail to reject H0

Interpretation

Since the p-value (0.897) is greater than the significance level (0.05), the null hypothesis is not rejected.

The result does not provide sufficient statistical evidence of a difference between the mean Total_Sales of the North and South regions at the 5% significance level.

4.3 One-Way ANOVA

A one-way ANOVA was performed to compare mean Total_Sales across the four regions: East, North, South, and West.

Results:

- Groups = East, North, South, West

- F-statistic = 2.164
- p-value = 0.097
- Significance level = 0.050
- Decision = Fail to reject H0

Interpretation

Since the p-value (0.097) is greater than the significance level (0.05), the null hypothesis is not rejected.

The result does not provide sufficient statistical evidence of a difference in mean Total_Sales across the four regions at the 5% significance level.

5. Summary of Hypothesis Tests

Test	Statistic	p-value	Decision
One-Sample T-Test	2.565	0.012	Reject H0
Welch T-Test: North vs South	0.130	0.897	Fail to reject H0
One-Way ANOVA	2.164	0.097	Fail to reject H0

6. Key Findings

1. The one-sample t-test indicates that the population mean of Total_Sales differs statistically from the benchmark value of 97,955.500.
2. The Welch t-test does not provide sufficient statistical evidence of a difference between the mean sales of the North and South regions.
3. The one-way ANOVA does not provide sufficient statistical evidence of a difference in mean sales across the four regions.
4. Statistical significance should be interpreted using the selected significance level of 0.05 rather than relying only on observed differences in sample means.

7. Business Insights

The statistical analysis provides several useful insights for business decision-making.

The significant result from the one-sample t-test suggests that the observed overall sales level differs from the selected benchmark. This can be useful for evaluating business performance against a predefined target or reference value.

The regional tests did not provide sufficient statistical evidence of differences in mean sales at the 5% significance level. Therefore, observed differences between regional sales should be interpreted cautiously and should not automatically be considered statistically meaningful.

Businesses can use these findings together with additional factors such as product performance, customer behavior, sales trends, and operational costs when evaluating regional performance.

8. Conclusion

This project demonstrates the application of statistical analysis and hypothesis testing to business sales data.

Three statistical tests were conducted: a one-sample t-test, an independent-samples Welch t-test, and a one-way ANOVA.

The one-sample t-test resulted in rejection of the null hypothesis, while the Welch t-test and one-way ANOVA resulted in failure to reject their respective null hypotheses at the 5% significance level.

Overall, the analysis demonstrates how statistical techniques can be used to distinguish statistically significant findings from observed differences in business data.

Note: Failing to reject a null hypothesis does not prove that the null hypothesis is true; it means that the test did not provide sufficient evidence against it at the selected significance level.